

secure log-in, collect statistics to optimize site functionality, and deliver content tailored to your interests. We want you to have control over the

More 



a [cookie consent tool](#) to set your own preferences. Otherwise, if you agree to our use of

more about our use of cookies, read our [Cookie Notice](#).



Learn ML with our
free downloadable
guide



nc blogs 

Download Now

Machine Learning & Big Data Blog

Anomaly Detection with Machine Learning: An Introduction



September 16, 2020

More

6 minute read



Learn ML with our
free downloadable
guide

[Jonathan Johnson](#)

is an instrumental role in robust distributed
anomaly detection can:

communication around system behavior

root cause analysis

integration to the software ecosystem

[Download Now](#)

anomaly detection is manual. However, [machine learning](#) is improving the success of anomaly detectors. Of course, [machine learning](#), there are upfront costs—data requirements and engineering talent.

In this article, we'll take a look at:

[Anomaly detection](#)[Machine learning AD](#)[Benchmarks](#)[How to begin](#)

What is anomaly detection?

Anomaly detection is any process that finds the outliers of a dataset; those items that don't belong. These anomalies might point to unusual network traffic, uncover a sensor on the fritz, or simply identify [data for cleaning](#), before analysis.

In today's world of distributed systems, managing and monitoring the system's performance is a chore—albeit a necessary chore. With hundreds or thousands of items to watch, anomaly detection can help point out where an error is occurring, enhancing root cause analysis and quickly getting tech support on the issue. Anomaly detection helps



Learn ML with our
free downloadable
guide



Download Now

ly detection is commonly used for:

n

onitoring

i sensor networks

ances

The challenge of anomaly detection

But even in these common use cases, above, there are some drawbacks to anomaly detection. From a conference paper [by Bram Steenwinckel](#):

“Anomaly detection (AD) systems are either manually built by experts setting thresholds on data or constructed automatically by learning from the available data through machine learning (ML).”

It is tedious to build an anomaly detection system by hand. This requires domain knowledge and—even more difficult to access—foresight.

For an ecosystem where the data changes over time, like fraud, this cannot be a good solution. Building a wall to keep out people works until they find a way to go over, under, or around it. When the system fails, builders need to go back in, and manually add further security methods.

Under the lens of chaos engineering, manually building anomaly detection is bad because it creates a system that cannot adapt (or is costly and untimely to adapt).

Machine learning then suits the engineer's purpose to create an AD

Learn ML with our
free downloadable
guide

n time

assets

anomaly detection with machine learning can
n conditions.



Download Now

Machine learning to anomaly detection requires a good
understanding of the problem, especially in situations with
unstructured data.

Structured data already implies an understanding of the problem
space. Anomalous data may be easy to identify because it breaks
certain rules. If a sensor should never read 300 degrees Fahrenheit and
the data shows the sensor reading 300 degrees Fahrenheit—there's
your anomaly. There is a clear threshold that has been broken.

Fraud detection in the early anomaly algorithms could work because
the data carried with it meaning. The data came structured, meaning
people had already created an **interpretable setting** for collecting data.
Their data carried significance, so it was possible to create random
trees and look for fraud.

However, **dark data** and unstructured data, such as images encoded as
a sequence of pixels or **language encoded as a sequence of characters**,
carry with it little interpretation and render the old algorithms useless...
until the data becomes structured. Structure can be found in the last
layers of a convolutional neural network (CNN) or in any number of
sorting algorithms.

Large datasets needed

machine learning model is trained requires dataset. The only way there is no data to support the claim, then the claim cannot hold in court.

Learn ML with our free downloadable guide



Download Now

res datasets; inferences can be made only when lated. Anomaly detection benefits from even because the assumption is that anomalies are

ly occur in the presence of abundance.

ed

...ing engineers are necessary. Obvious, but overlooked. Machine learning talent is not a commodity, repair shops, not all engineers are equal.

It requires skill and craft to build a good Machine Learning model. The cost to get an anomaly detector from 95% detection to 98% detection could be a few years and a few ML hires.

Anomaly detection in three settings

In a 2018 lecture, [Dr. Thomas Dietterich](#) and his team at Oregon State University explain how anomaly detection will occur under three different settings. They all depend on the condition of the data. The three settings are:

- 1. Supervised
- 2. Clean
- 3. Unsupervised

Anomaly Detection: Algorithms, Explanations, Appl...



ore ▾



Learn ML with our
free downloadable
guide



g in depth.

ed with “nominal” or “anomaly”.

Download Now

ed setting is the ideal setting. It is the instance when a dataset comes neatly prepared for the data scientist with all data points labeled as anomaly or nominal. In this case, all anomalous points are known ahead of time. That means there are sets of data points that are anomalous, but are not identified as such for the model to train on.

Popular ML algorithms for structured data:

Support vector machine learning

k-nearest neighbors (KNN)

Bayesian networks

Decision trees

2. Clean

In the Clean setting, all data are assumed to be “nominal”, and it is contaminated with “anomaly” points.

The clean setting is a less-ideal case where a bunch of data is presented to the modeler, and it is clean and complete, but all data are presumed to be nominal data points. Then, it is up to the modeler to detect the anomalies inside of this dataset.

In Unsupervised settings, the training data is unlabeled and consists of

"y" points.

Learn ML with our
free downloadable
guide



Download Now

The ever-increasing case for modelers in the
case of **dark data**, is the unsupervised instance.
Unsupervised case do not have their parts labeled
for us. There is no ground truth from which to
derive. The model must show the modeler what is
nominal.

*tasks within unsupervised learning are clustering,
dimensionality reduction, and density estimation. In all of these cases,
we wish to learn the inherent structure of our data without using
explicitly-provided labels."- Devin Soni*

In the Unsupervised setting, a different set of tools are needed to
create order in the unstructured data. In unstructured data, the
primary goal is to create clusters out of the data, then find the few
groups that don't belong. Really, all anomaly detection algorithms are
some form of **approximate density estimation**.

Popular ML Algorithms for unstructured data are:

- Self-organizing maps (SOM)
- K-means
- C-means
- Expectation-maximization meta-algorithm (EM)
- Adaptive resonance theory (ART)
- One-class support vector machine

Learn ML with our
free downloadable
guide



Download Now

K-Means Clustering ([Source](#))

ecture slides ([PDF](#)), the strategies for anomaly
of the unsupervised setting are broken down into

Where machine learning isn't appropriate, top non-ML detection
algorithms include:

IFOR: Isolation Forest (Liu, et al., 2008)

LODA: Lightweight Online Detector of Anomalies (Pevny, 2016)

detection

More ▾



Learn ML with our
free downloadable
guide



[Anomaly Benchmark](#). From the GitHub Repo:

[Download Now](#)

“NAB is a novel benchmark for evaluating algorithms for anomaly detection in streaming, real-time applications. It is composed of over 50 labeled real-world and artificial time series data files plus a novel scoring mechanism designed for real-time applications.”

Thus far, on the NAB benchmarks, the best performing anomaly detector algorithm catches 70% of anomalies from a real-time dataset.

Get started with ML anomaly detection

If you want to get started with machine learning anomaly detection, I suggest started here:

[Anomaly Detection Techniques in Python](#) (Christopher Jose)

Learn ML with our free downloadable guide

[Anomaly Detection with scikit-learn](#)
[Machine Learning](#)

resources

Related topics, explore these resources:



Download Now

Learn Guide

[Python vs Java: Why Python is Becoming More Popular than Java](#)

Learn ML with our free downloadable guide

This e-book teaches machine learning in the simplest way possible. This book is for managers, programmers, directors – and anyone else who wants to learn machine learning. We start with very basic stats and algebra and build upon that.



Download Now

These postings are my own and do not necessarily represent BMC's position, strategies, or opinion.

See an error or have a suggestion? Please let us know by emailing blogs@bmc.com.



Learn ML with our
free downloadable
guide



Download Now

MC Bring the A-Game

IC delivers the software and services that enable nearly
ling 84% of the Forbes Global 100, to thrive in their
omous Digital Enterprise.

You may also like

Outlier and Anomaly
Detection with Machine
Learning

MongoDB vs Cassandra:
NoSQL Databases Compared

Reduce MTTR: Machine
Learning to the Rescue



Structured vs Unstructured
Data: A Shift in Privacy

Creating & Using Linked
Tables in Power BI

Hadoop Careers: Getting
Hired



Learn ML with our
free downloadable
guide



Download Now

About the author

Johnson

Johnson is a tech writer who integrates life and technology. He is focused on helping people's degrees of freedom. Visit his website at [johnsonm.com](#).



Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

Learn ML with our
free downloadable
guide



Download Now

By using this site signifies your acceptance of BMC's **Terms of Use**. BMC, the BMC logo, and other BMC marks are registered and may be registered in the U.S. and in other countries.